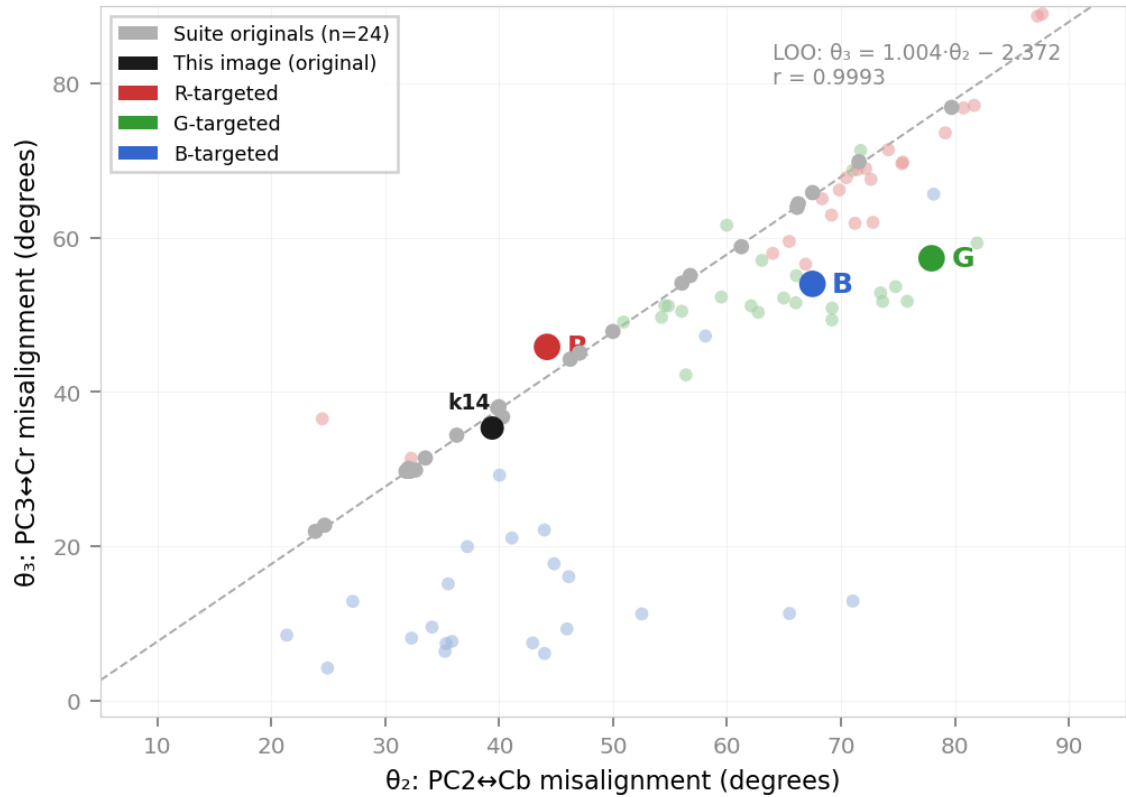


Compression Profile: KODIM14 768x512 | 24-bit RGB | PNG (lossless)

white water rafters | Two-Dimensional (PC1 = 81.6%)

1. Geometric Displacement Under Directional Perturbation



2. Pairwise Correlation and Angular Misalignment

Version	R-G	R-B	G-B	Avg r	theta2	theta3	L00 dev
Original	0.8613	0.4568	0.7107	0.676	39.38°	35.34°	-1.83°
R-targeted	-0.0554	-0.2490	0.9011	0.402	44.2°	45.86°	+3.86°
G-targeted	-0.0365	0.9502	-0.0303	0.339	77.96°	57.35°	-18.55°
B-targeted	0.9242	-0.1211	0.1108	0.385	67.5°	54.03°	-11.37°

R-targeted isolates red (R-G, R-B < 0.25) while preserving G-B coupling at 0.901.

G-targeted isolates green (R-G, G-B < 0.04) while preserving R-B at 0.950.

B-targeted preserves R-G at 0.924 while driving R-B and G-B near zero.

All three perturbation directions depart from the L00 constraint manifold ($r = 0.9993$).

Compression Profile: KODIM14768x512 | 24-bit
RGB | PNG (lossless)

white water rafters | Two-Dimensional (PC1 = 81.6%)

3. File Format and Size Comparison

All versions depict the same scene. BPP = (file size in bits) / (pixel count).

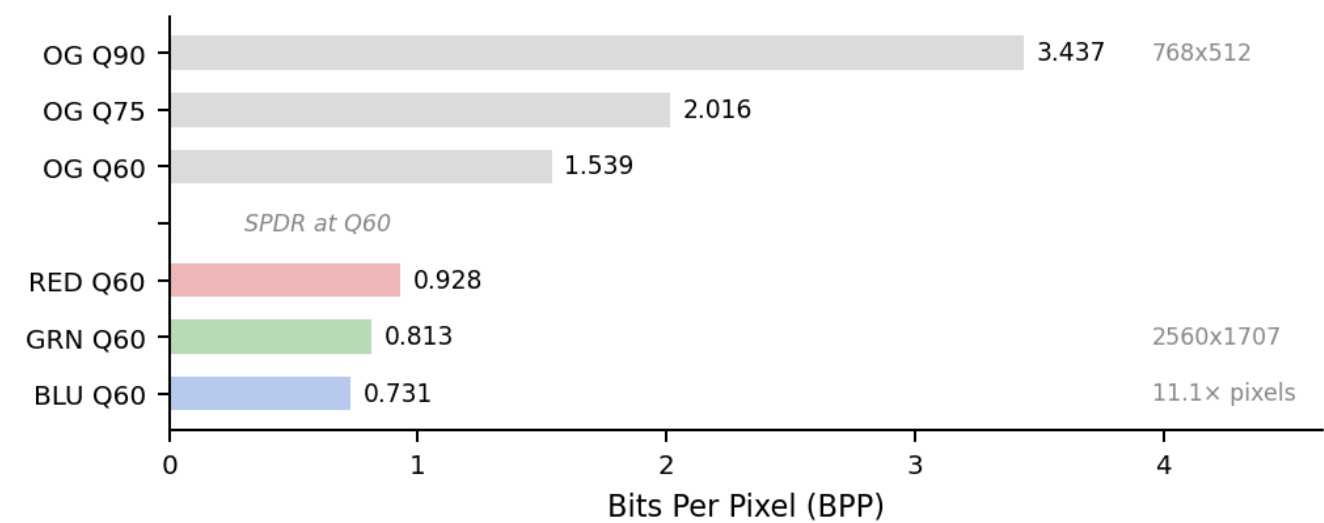
ORIGINAL (kodim14)

Version	Format	Resolution	Size	BPP	Pixels
PNG (lossless)	PNG	768x512	676.0 KB	14.083	393,216
JPEG Q90	JPEG	768x512	165.0 KB	3.437	393,216
JPEG Q75	JPEG	768x512	96.8 KB	2.016	393,216
JPEG Q60	JPEG	768x512	73.9 KB	1.539	393,216

REDISTRIBUTED (kodim14 SPDR at Q60)

Version	Format	Resolution	Size	BPP	Pixels
RED Q60	JPEG	2560x1707	495.2 KB	0.928	4,369,920
GRN Q60	JPEG	2560x1707	433.7 KB	0.813	4,369,920
BLU Q60	JPEG	2560x1707	389.8 KB	0.731	4,369,920

4. Quality Ladder: SPDR Q60 vs Original Q90



All three perturbation directions at Q60 produced lower BPP than the original at Q90, while carrying 11.1× the original pixel count.

Compression Profile: KODIM14 768x512 | 24-bit RGB | PNG (lossless)

white water rafters | Two-Dimensional (PC1 = 81.6%)

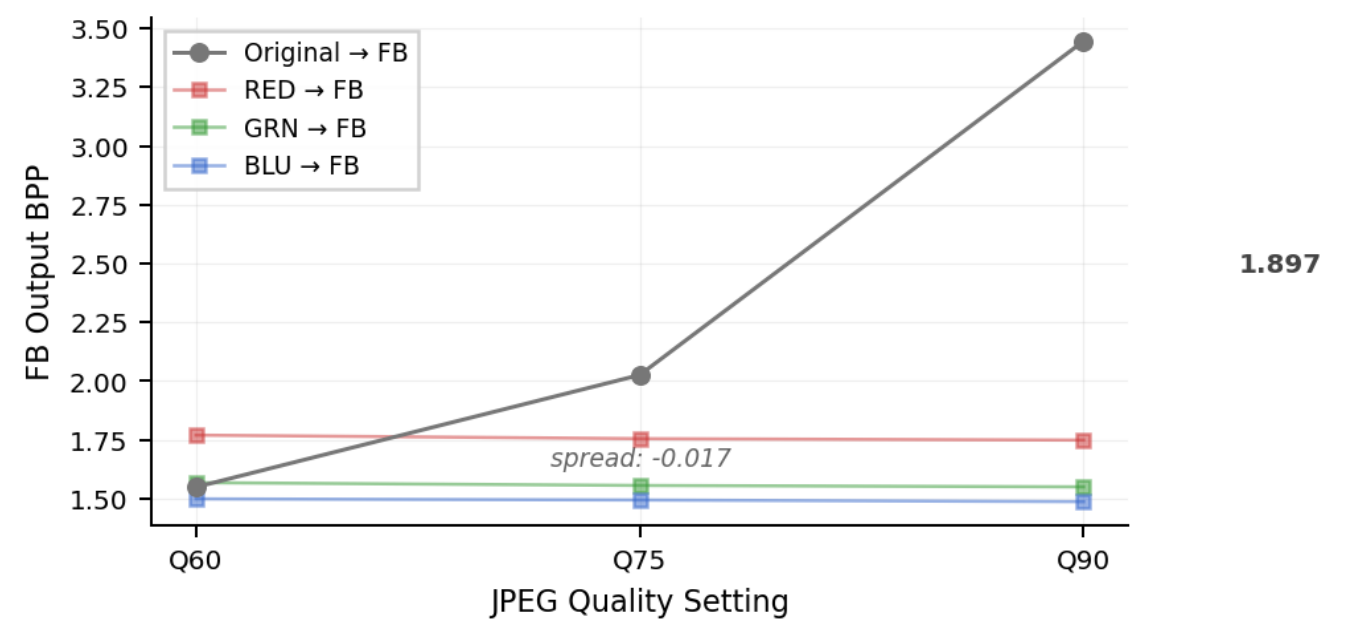
5. Third-Party Compression Pipeline (Facebook/Meta)

Each version uploaded to Facebook. Platform-compressed output measured at steady state (FB2).

Pipeline	Input Res	Output Res	FB BPP	BPP reduction
OG PNG → FB	768x512	768x512	3.218	—
R-targeted → FB	2560x1707	2048x1366	1.749	45.6%
G-targeted → FB	2560x1707	2048x1366	1.550	51.8%
B-targeted → FB	2560x1707	2048x1366	1.487	53.8%
ABC (no perturbation) → FB	2560x1707	2048x1366	1.395	56.7%

ABC control: same pipeline, same resolution, no covariance perturbation. Avg |r| unchanged (0.6737 vs 0.676 original). Comparable BPP reduction was not observed in the ABC control.

6. Quality Parameter Sensitivity Through Facebook



Original: Q60 → Q90 produces a 1.897 BPP spread through Facebook.
Redistributed: Q60 → Q90 produces a 0.017 BPP spread through Facebook.
Redistributed versions produced substantially smaller Q60-90 BPP variation through the measured pipeline.

7. Pipeline Summary – kodim14

Best BPP (Q60, direct)	0.731	2560x1707, 4,369,920 px
Best BPP through FB	1.487	2048x1366 output
vs OG Q90 → FB	56.8%	SPDR Q60→FB vs OG Q90→FB
Pixel ratio	11.1×	4,369,920 vs 393,216